Math 101 Extra Credit

Show complete handwritten work for all 50 questions on separate paper. Answers are in red; answers alone do not earn credit.

1. Solve for x: $5(2x-3)-4=3(x+2)+10$.

Answer: $x=5$

2. Solve for x: $4-3(2x+5)=2(x-7)-5$.

Answer: $x=1$

3. Solve for x: $7(x-2)+3=4(2x+1)-9$.

Answer: $x=-6$

4. Solve for x: $2(3x+4)-5(x-2)=31$.

Answer: $x=13$

5. Solve for x: $\dfrac{x+2}{3}-\dfrac{x-4}{5}=4$.

Answer: $x=19$

6. Solve for t: $\dfrac{2t-1}{4}+\dfrac{t+3}{6}=5$.

Answer: $t=\frac{57}{8}$

7. Solve for p: $\dfrac{p-2}{3}-\dfrac{2p+1}{4}=-2$.

Answer: $p=\frac{13}{2}$

8. Solve for s: $\dfrac{3s+2}{5}-\dfrac{s-1}{2}=s-4$.

Answer: $s=\frac{49}{9}$

9. Find the equation of the line through $(-3,7)$ and $(5,-9)$ in slope-intercept form.

Answer: $y=-2x+1$

10. Find the equation of the line through $(-2,-5)$ and $(4,4)$ in slope-intercept form.

Answer: $y=\frac{3}{2}x-2$

11. Find the equation of the line through $(2,-1)$ and $(6,1)$ in slope-intercept form.

Answer: $y=\frac{1}{2}x-2$

12. Find the equation of the line through $(-4,-6)$ and $(2,-6)$ in slope-intercept form.

Answer: $y=-6$

13. Find the equation of the line parallel to $2x-3y=9$ through $(3,-1)$. Use slope-intercept form.

Answer: $y=\frac{2}{3}x-3$

14. Find the equation of the line perpendicular to $y=-\frac{3}{4}x+2$ through $(-6,1)$. Use slope-intercept form.

Answer: $y=\frac{4}{3}x+9$

15. Find the equation of the line parallel to $y=-\frac{5}{2}x+7$ through $(4,1)$. Use slope-intercept form.

Answer: $y=-\frac{5}{2}x+11$

16. Find the equation of the line perpendicular to $4x+2y=10$ through $(-2,5)$. Use slope-intercept form.

Answer: $y=\frac{1}{2}x+6$

17. Solve the system: $\begin{cases}2x+y=11\\x-y=1\end{cases}$

Answer: $(x,y)=(4,3)$

18. Solve the system: $\begin{cases}3x-2y=4\\5x+y=11\end{cases}$

Answer: $(x,y)=(2,1)$

19. Solve the system: $\begin{cases}4x+y=-5\\2x-3y=15\end{cases}$

Answer: $(x,y)=(0,-5)$

20. Solve the system: $\begin{cases}2x+5y=9\\3x-2y=-15\end{cases}$

Answer: $(x,y)=(-3,3)$

21. Solve the system: $\begin{cases}3x+4y=5\\5x-2y=17\end{cases}$

Answer: $(x,y)=(3,-1)$

22. Solve the system: $\begin{cases}x+2y=0\\3x-4y=10\end{cases}$

Answer: $(x,y)=(2,-1)$

23. A student has test scores of 72, 81, 76, 79. The final exam counts as 2 test grades. Write a linear equation for course grade y in terms of final-exam grade x, then find the final-exam grade needed for a 80% course average.

Answer: $y=\frac{308+2x}{6}$; final-exam grade: 86%.

24. A student has test scores of 64, 78, 82, 76. The final exam counts as 3 test grades. Write a linear equation for course grade y in terms of final-exam grade x, then find the final-exam grade needed for a 82% course average.

Answer: $y=\frac{300+3x}{7}$; final-exam grade: $\frac{274}{3}$%.

25. A student has test scores of 88, 79, 83, 86. The final exam counts as 4 test grades. Write a linear equation for course grade y in terms of final-exam grade x, then find the final-exam grade needed for a 88% course average.

Answer: $y=\frac{336+4x}{8}$; final-exam grade: 92%.

26. A rectangular study room is 7 meters longer than it is wide. Its perimeter is 54 meters. Find its length and width.

Answer: Length: 17 m; width: 10 m.

27. An employee receives a fixed monthly salary and a year-end bonus of \$4,800. Total annual earnings are \$78,600. Find the monthly salary.

Answer: \$6,150 per month.

28. A school buys notebooks at \$4 each and packs of folders at \$7 each. It buys 23 items and spends \$125. How many notebooks and how many packs of folders does it buy?

Answer: 12 notebooks; 11 packs of folders.

29. A bag contains 48 blue and green marbles. The number of green marbles is 3 more than twice the number of blue marbles. How many marbles of each color are in the bag?

Answer: 15 blue marbles; 33 green marbles.

30. A cash drawer contains only ten-dollar and twenty-dollar bills. There are three times as many ten-dollar bills as twenty-dollar bills, and the total is \$1,000. Find the number of each type of bill.

Answer: 60 ten-dollar bills; 20 twenty-dollar bills.

31. A lab mixes a 15% acid solution and a 45% acid solution to make 120 mL of a 30% solution. How many milliliters of each solution are needed?

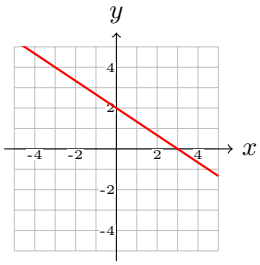
Answer: 60 mL of the 15% solution; 60 mL of the 45% solution.

32. A store mixes peanuts costing \$5 per pound with almonds costing \$8 per pound. How many pounds of each are needed for 40 pounds of a mixture costing \$6.20 per pound?

Answer: 24 lb of peanuts; 16 lb of almonds.

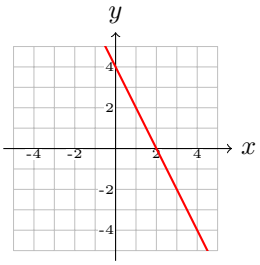
33. Graph $2x + 3y = 6$. Show your calculations for both intercepts.

Answer: $y = -\frac{2}{3}x + 2$; intercepts: $(0, 2)$ and $(3, 0)$.



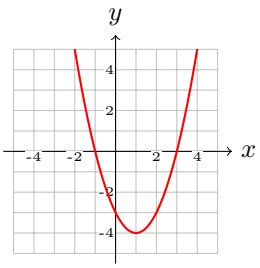
34. Graph $y = -2x + 4$. Show your calculations for both intercepts.

Answer: Intercepts: $(0, 4)$ and $(2, 0)$.



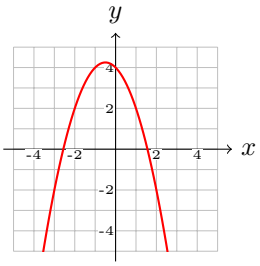
35. Graph $y = x^2 - 2x - 3$ by making a table for $x = -2, -1, 0, 1, 2$.

Answer: Points: $(-2, 5)$, $(-1, 0)$, $(0, -3)$, $(1, -4)$, $(2, -3)$.

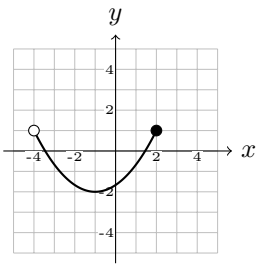


36. Graph $y = -x^2 - x + 4$ by making a table for $x = -2, -1, 0, 1, 2$.

Answer: Points: $(-2, 2)$, $(-1, 4)$, $(0, 4)$, $(1, 2)$, $(2, -2)$.

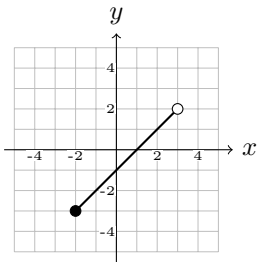


37. Write the domain of the displayed function in interval notation.



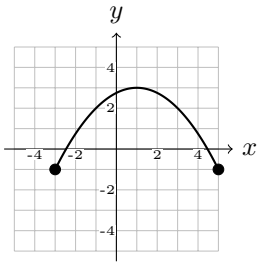
Answer: $(-4, 2]$

38. Write the domain of the displayed function in interval notation.



Answer: $[-2, 3)$

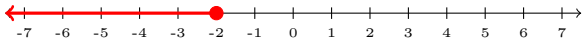
39. Write the domain of the displayed function in interval notation.



Answer: $[-3, 5]$

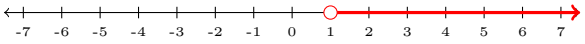
40. Graph $x \leq -2$ on a number line and write the solution in interval notation.

Answer: $(-\infty, -2]$



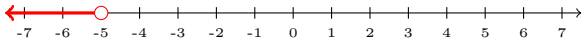
41. Graph $x > 1$ on a number line and write the solution in interval notation.

Answer: $(1, \infty)$



42. Graph $x < -5$ on a number line and write the solution in interval notation.

Answer: $(-\infty, -5)$



43. Simplify completely using positive exponents. Assume all variables are nonzero.

$$\left(\frac{x^{-2}y^3}{xy^{-1}}\right)^{-2}.$$

Answer: $\dfrac{x^6}{y^8}$

44. Simplify completely using positive exponents. Assume all variables are nonzero.

$$\left(\frac{a^3b^{-2}}{a^{-1}b}\right)^{2}.$$

Answer: $\dfrac{a^8}{b^6}$

45. Simplify completely using positive exponents. Assume all variables are nonzero.

$$\left(\frac{x^{-3}y^2z}{x^{-1}y^{-2}z^3}\right)^{-1}.$$

Answer: $\dfrac{x^2z^2}{y^4}$

46. Simplify completely using positive exponents. Assume all variables are nonzero.

$$\left(\frac{m^2n^{-3}}{m^{-1}n^{-1}}\right)^{-3}.$$

Answer: $\dfrac{n^6}{m^9}$

47. Let $f(x) = 2x^3 + 3x^2 - 4x + 5$. Find $f(-2)$.

Answer: 9

48. Let $g(x) = -x^3 + 2x^2 + 5x - 6$. Find $g(3)$.

Answer: 0

49. Write 4.073×10^5 as a whole number or decimal.

Answer: 407,300

50. Evaluate $-|-23|$.

Answer: -23